

Runbook: MCO User Lock / Restore

Item	Value
Script	Set-McoUserLock.ps1
Purpose	Temporarily disable all active MCO users (option 1) and restore them to their original values afterwards (option 2)
Database	Oracle 11g, host 10.152.218.205 , port 1526 , SID MCODEV
Table	MC0001.UTILISATEUR
Excluded users (never changed)	PATKE01 , CICAM01 , KHARI01
Credential file	MCODEV.cred (next to the script)
Backup file	UTILISATEUR_locked_users.csv (next to the script)
Runtime	Windows PowerShell 5.1

1. What the script does

1.1 Option 1 — Disable active users

An **active user** is one where `NBRE_TENTATIVE = 0` **and** `COMPTE_VEROUILLE = 0`.

Step	Action
1	Checks that the backup file does not already exist. If it does, the script stops, so original values are never overwritten
2	Connects to 10.152.218.205:1526/MCODEV and starts a transaction
3	Selects and locks (<code>FOR UPDATE</code>) every active user except the excluded codes
4	Lists the users and asks Disable N user(s) listed above? (default No)
5	On Yes : writes the original values to the backup CSV before saving any change
6	Sets <code>NBRE_TENTATIVE = 3</code> and <code>COMPTE_VEROUILLE = 1</code> for each listed user
7	Commits only if the number of rows updated matches the number listed; otherwise rolls back and deletes the backup

```
-- Selection
SELECT CODE_UTILISATEUR, NOM_UTILISATEUR, PRENOM_UTILISATEUR, NBRE_TENTATIVE, COMPTE_VEROUILLE
FROM MC0001.UTILISATEUR
WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0
AND UPPER(TRIM(CODE_UTILISATEUR)) NOT IN ('PATKE01', 'CICAM01', 'KHARI01')
FOR UPDATE;

-- Update (per user)
UPDATE MC0001.UTILISATEUR
SET NBRE_TENTATIVE = 3, COMPTE_VEROUILLE = 1
WHERE CODE_UTILISATEUR = :code AND NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0;
```

1.2 Option 2 — Restore users

Step	Action
1	Checks that the backup file exists
2	Lists the users in the backup and asks Restore N user(s) listed above to their original values? (default No)
3	For each user, sets <code>NBRE_TENTATIVE</code> and <code>COMPTE_VEROUILLE</code> back to their original values, only if they still hold <code>3</code> and <code>1</code>
4	Users changed by someone else since the lock are skipped and listed in a warning
5	Commits all restores in one transaction
6	Renames the backup to <code>UTILISATEUR_locked_users.restored_yyyyMMdd_HHmss.csv</code> so it can't be applied twice

```
UPDATE MC0001.UTILISATEUR
SET NBRE_TENTATIVE = :origN, COMPTE_VEROUILLE = :origC
WHERE CODE_UTILISATEUR = :code AND NBRE_TENTATIVE = 3 AND COMPTE_VEROUILLE = 1;
```

1.3 Backup file format

CSV with a header row, UTF-8. One row per user changed by option 1.

Column	Content
<code>CODE_UTILISATEUR</code>	User code
<code>NOM_UTILISATEUR</code> / <code>PRENOM_UTILISATEUR</code>	Last name / first name
<code>ORIG_NBRE_TENTATIVE</code> / <code>NEW_NBRE_TENTATIVE</code>	Value before (<code>0</code>) / after (<code>3</code>)
<code>NBRE_TENTATIVE_CHG</code>	Y if the column was changed
<code>ORIG_COMPTE_VEROUILLE</code> / <code>NEW_COMPTE_VEROUILLE</code>	Value before (<code>0</code>) / after (<code>1</code>)
<code>COMPTE_VEROUILLE_CHG</code>	Y if the column was changed
<code>CHANGED_AT</code>	Timestamp of the lock run

Do not edit, move or delete the backup file between option 1 and option 2. It is the only record of the original values.

2. Prerequisites

#	Requirement	How to check
1	Windows PowerShell 5.1	<code>\$PSVersionTable.PSVersion</code>
2	Network access to 10.152.218.205 on port 1526	<code>Test-NetConnection 10.152.218.205 -Port 1526</code> , which should return <code>TcpTestSucceeded : True</code>
3	Oracle account with <code>SELECT</code> and <code>UPDATE</code> on <code>MC0001.UTILISATEUR</code>	The same login can run an <code>UPDATE</code> in SQL Developer
4	<code>Oracle.ManagedDataAccess.dll</code> version 19.18.0 in the same folder as the script	See section 3.1

3. One-time setup

3.1 Install the Oracle driver (19.18.0)

Skip this step if the DLL is already in the folder (e.g. from the AGPH extract setup).

```
cd $env:USERPROFILE\Downloads
[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
Invoke-WebRequest https://www.nuget.org/api/v2/package/Oracle.ManagedDataAccess/19.18.0 -OutFile odp.zip
Expand-Archive odp.zip -DestinationPath odp -Force
Copy-Item .\odp\lib\net40\Oracle.ManagedDataAccess.dll .
Unblock-File .\Oracle.ManagedDataAccess.dll
Unblock-File .\Set-McoUserLock.ps1
```

Use 19.18.0, not 23.x. Version 23.x needs extra .NET files that aren't on a standard Windows PC.

3.2 Save the Oracle credential

The script handles this itself. On the first run it asks for the Oracle user name and password and saves them to `MCODEV.cred`. Later runs load the file without prompting.

Rule	Reason
Run the script as the Windows user who will use it from now on	The file is encrypted with Windows DPAPI: only that user, on that machine, can decrypt it

Rule	Reason
Run with <code>-ResetCred</code> whenever the Oracle password changes	Otherwise you get <code>ORA-01017</code>
Never copy the <code>.cred</code> file to another PC or user	It won't decrypt there

4. Run procedure

4.1 Pre-checks (before option 1)

Check	Action
Change approved	Confirm the change ticket / approval to lock users
No leftover backup	Test-Path <code>.\UTILISATEUR_locked_users.csv</code> must return <code>False</code>
Expected volume	In SQL Developer: <code>SELECT COUNT(*) FROM MC0001.UTILISATEUR WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0 AND CODE_UTILISATEUR NOT IN ('PATKE01', 'CICAM01', 'KHARI01');</code> — note the number
Users informed	Users have been told they will be unable to log in

4.2 Option 1 — Disable active users

```
cd $env:USERPROFILE\Downloads
.\Set-McoUserLock.ps1 -Option 1
```

Expected output:

```
Using credentials for MC0001 from C:\Users\<you>\Downloads\MCODEV.cred
Connected to 10.152.218.205:1526/MCODEV

CODE_UTILISATEUR NOM_UTILISATEUR PRENOM_UTILISATEUR ORIG_NBRE_TENTATIVE ORIG_COMPTE_VEROUILLE
-----
ABCDE01          ...           ...                               0                               0
...

Confirm
Disable 157 user(s) listed above?
[Y] Yes [N] No [?] Help (default is "N"): Y
Disabled 157 user(s). Originals saved to C:\Users\<you>\Downloads\UTILISATEUR_locked_users.csv
```

The count must match the pre-check count in 4.1. If it doesn't, answer **N** and investigate.

4.3 Option 2 — Restore users

```
cd $env:USERPROFILE\Downloads
.\Set-McoUserLock.ps1 -Option 2
```

Expected output:

```
Using credentials for MC0001 from C:\Users\<you>\Downloads\MCODEV.cred
Connected to 10.152.218.205:1526/MCODEV
...
Restore 157 user(s) listed above to their original values?
[Y] Yes  [N] No  [?] Help (default is "N"): Y
Restored 157 user(s).
Backup archived to C:\Users\<you>\Downloads\UTILISATEUR_locked_users.restored_20261006_153012.csv
```

If some users were changed after the lock, you also see:

```
WARNING: Skipped 2 user(s) - not found or values changed since lock: ABCDE01, FGHIJ02
```

Review those users manually (section 5).

4.4 Interactive menu

Running the script with no `-Option` shows a menu:

```
MC0001.UTILISATEUR on 10.152.218.205:1526/MCODEV
1) Disable all active users (except PATKE01, CICAM01, KHARI01)
2) Restore users from C:\Users\<you>\Downloads\UTILISATEUR_locked_users.csv

Select option (1/2):
```

4.5 Parameters

Parameter	Default	Use
<code>-Option</code>	<i>(menu)</i>	<code>1</code> = disable, <code>2</code> = restore
<code>-CredFile</code>	<code><script folder>\MCODEV.cred</code>	Saved credential file
<code>-ResetCred</code>	Off	Prompt again and overwrite the <code>.cred</code> file
<code>-UserName</code>	<i>(prompt)</i>	Oracle user name used when creating the <code>.cred</code> file
<code>-Credential</code>	<i>(from .cred)</i>	Pass a credential directly; bypasses the <code>.cred</code> file
<code>-BackupFile</code>	<code><script folder>\UTILISATEUR_locked_users.csv</code>	Where option 1 writes and option 2 reads the original values
<code>-ExcludeUsers</code>	<code>PATKE01, CICAM01, KHARI01</code>	User codes never changed by option 1
<code>-DbHost</code>	<code>10.152.218.205</code>	Database host

Parameter	Default	Use
-Port	1526	Listener port
-Sid	MCODEV	SID, or the service name when used with -UseServiceName
-UseServiceName	Off	Connect with SERVICE_NAME instead of SID
-Schema	MC0001	Schema owning UTILISATEUR
-DllPath	<script folder>\Oracle.ManagedDataAccess.dll	Driver location

5. Post-run verification

After	Check (SQL Developer)	Pass criteria
Option 1	SELECT COUNT(*) FROM MC0001.UTILISATEUR WHERE NBRE_TENTATIVE = 0 AND COMPTE_VEROUILLE = 0;	Only the excluded users remain (≤ 3)
Option 1	SELECT CODE_UTILISATEUR, NBRE_TENTATIVE, COMPTE_VEROUILLE FROM MC0001.UTILISATEUR WHERE CODE_UTILISATEUR IN ('PATKE01', 'CICAM01', 'KHARI01');	Unchanged
Option 1	Backup file	Exists; row count (excluding header) equals the disabled count
Option 2	Same COUNT(*) as option 1	Equals the pre-check count from 4.1 plus the excluded active users
Option 2	Skipped users (if any)	Each one reviewed and fixed manually or confirmed as intended
Option 2	Backup file	Renamed to *.restored_<timestamp>.csv

6. Exit codes

Code	Meaning
0	Finished: changes committed, cancelled at the prompt, or nothing to do
1	Failed. The error is printed (e.g. Failed: ORA-...). All changes in that run are rolled back

7. Troubleshooting

Symptom	Cause	Fix
Backup file '...' already exists (option 1)	A previous lock has not been restored	Run option 2 first, or use <code>-BackupFile</code> with a new name if the old file is no longer needed
Backup file '...' not found (option 2)	Wrong folder, or already restored	Run from the folder holding the CSV, or pass <code>-BackupFile</code> . Check for a <code>*.restored_*.csv</code>
ORA-00932: inconsistent datatypes: expected CHAR got NUMBER (option 2)	Old script version (before 6 Oct 2026)	Download the current script and run option 2 again. Nothing was changed by the failed run
Script waits after Connected to ...	Another session holds uncommitted changes on UTILISATEUR rows	Ask the other session (or KYNDRYL Oracle Database Support) to commit or roll back; or press Ctrl+C and retry later
Skipped N user(s) ... (option 2)	Those users were changed after the lock (e.g. reset by support)	Check each user in SQL Developer and fix manually if needed
Expected N rows updated, got M. Rolled back.	Rows changed between select and update	Re-run option 1; nothing was saved
Cannot read '...MCODEV.cred' / Key not valid for use in specified state	<code>.cred</code> created by another Windows user or PC	Run with <code>-ResetCred</code>
ORA-01017: invalid username/password	Wrong or changed password	Run with <code>-ResetCred</code>
ORA-28000: account is locked	Oracle account locked	Ask KYNDRYL Oracle Database Support to unlock it
ORA-00942: table or view does not exist	No grant on MC0001.UTILISATEUR, or wrong schema	Ask KYNDRYL Oracle Database Support for <code>SELECT / UPDATE</code> ; check <code>-Schema</code>
ORA-01031: insufficient privileges	<code>SELECT</code> only, no <code>UPDATE</code>	Ask KYNDRYL Oracle Database Support for <code>UPDATE</code> on MC0001.UTILISATEUR
ORA-12505: listener does not currently know of SID	MCODEV is a service name, not a SID	Add <code>-UseServiceName</code>
ORA-12170 / ORA-12541 / timeout	Network or firewall	<code>Test-NetConnection 10.152.218.205 -Port 1526</code> ; check VPN; if blocked, submit a Firewall Request via ServiceNow
Oracle.ManagedDataAccess.dll not found	Driver missing	Section 3.1, or pass <code>-DllPath</code>
... running scripts is disabled on this system	Execution policy / downloaded file	<code>Unblock-File .\Set-McoUserLock.ps1</code> , or <code>powershell.exe -ExecutionPolicy Bypass -File .\Set-McoUserLock.ps1</code>

8. Security and housekeeping

Item	Guidance
Credentials	Only in the DPAPI-encrypted <code>MCODEV.cred</code> ; never in the script, command history or email
Backup CSV	Contains user codes and names. Keep it on a bank-managed drive; never commit it to Git (it is in <code>.gitignore</code>)
Restored backups	Keep <code>*.restored_*.csv</code> as the audit trail per retention policy, then delete
Excluded users	Change only with approval; pass <code>-ExcludeUsers</code> rather than editing the script
Script changes	Keep under version control (<code>ksolo13/fscprd</code>); test on MCODEV before any other environment

9. Contacts and escalation

Issue	Contact
Database access, grants, row locks	KYNDRYL — Oracle Database Support
Network or firewall to <code>10.152.218.205:1526</code>	Please submit a Firewall Request via ServiceNow
MCO application / user impact	Kevin Paterson
Script defects	Kevin Paterson

10. Change log

Date	Change
2026-10-06	Initial version: lock / restore with CSV backup
2026-10-06	Credentials saved in DPAPI-encrypted <code>MCODEV.cred</code> ; - <code>ResetCred</code> added
2026-10-06	Active user redefined as <code>NBRE_TENTATIVE = 0 and COMPTE_VEROUILLE = 0</code>
2026-10-06	Fixed <code>ORA-00932</code> on restore (plain <code>SET</code> instead of <code>CASE</code>)